# SIMATS ENGINEERING

## ASSESSMENT TOOL 1: Analytical Problem Solving

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

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**COURSE OUTCOME COVERED**

**CO2:** *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5
Duration : 60 Minutes

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**Code of Conduct**

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

**Assessment Tasks**

# Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

**Analytical Problem Solving Rubric**

| Criteria | Weightage | Level 4 - Exemplary | Level 3 – Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Problem Interpretation | 20% | Interprets all technical requirements accurately | Interprets most requirements correctly | Partial interpretation | Misinterprets the problem |
| Analytical Reasoning | 25% | Strong reasoning with clear cause-effect analysis | Good reasoning with minor gaps | Basic analysis only | Weak or no analysis |
| Method Selection | 20% | Chooses highly appropriate virtualization and security concepts | Chooses mostly suitable concepts | Partially suitable concepts | Inappropriate approach |
| Solution Quality | 20% | Provides feasible and technically sound solution | Reasonable solution with minor issues | Basic solution with limitations | Unclear or invalid solution |
| Presentation of Analysis | 15% | Well-structured and technically precise | Generally clear | Somewhat unclear | Difficult to follow |

**Total Marks Awarded**

**:**

1) A hypervisor is software that creates and Manages Virtual Machines (VMs).

Type-1 Hypervisor:

* Runs directly on physical hardware.
* Provides high Performance.
* Offers Strong isolation and Security.
* Suitable for cloud datacenters.
* Examples: VMWare ESXi, Microsoft Hyper-V.

Type-2 Hypervisor

* Runs on a host operating system.
* Has additional overhead.
* Security depends on the host OS and hypervisor.
* Suitable for testing and personal use.
* Examples: VMware Workstation, Oracle VirtualBox.

Analysis:

* Performance: Type-1 provides faster execution with less overhead.
* Security: Direct hardware accers reduces dependence on the host OS.
* Manageability: Supports Centralized management and Efficient resaurces allocation.

Conclusion:

Type-1 hypervisors is the better choice for mission-critical cloud applications because it provides better performance, security, reliability and scalability.

2) Virtualization allows finance, healthcare and student portal workloads to run on the same physical hardware while maintaining separate virtual environments.

How Virtualization supports isolation:

(i) VM isolation: Each workload runs in a separate virtual machine.

(ii) Memory isolation: Prevents one VM from accessing another VM's Memory.

(iii) Network isolation: Virtual networks and firewalls control communication between workloads.

(iv) Storage isolation: Separate virtual disks and access permissions protect sensitive data.

(v) Access Control: Role-based access control restricts unauthorized access.

Two major Attacks Surfaces:

- Hypervisor Vulnerabilities: Attackers may exploit the hypervisor to access multiple VMs.
- Virtual network vulnerabilities: Misconfigured virtual networks may allow unauthorized communication between workloads.

Security Measures:

* Regularly update and patch the hypervisor.
* Use network segmentation and firewalls.
* Apply strong Authentication and least-privilege access.
* Monitor VM activities and security logs.

Conclusion:

Proper virtualization and security controls provide effective workload isolation and protect sensitive cloud data.

3) Given data:

- CPU utilization : 92%
- Memory utilization : 88%
- Storage I/O latency : 35ms
- Average VM boot time : 170 seconds.

Analysis of Bottlenecks:

1) High CPU utilization (92%):
The host is heavily loaded, causing CPU contention and slower VM execution.

2) High memory utilization (88%):
Insufficient free memory may cause memory contention and swapping.

3) High storage I/O latency (35 ms):
Slow storage operations delay disk access and VM activities.

4) Long VM boot time (170 seconds):
High resource utilization and slow storage may increase VM startup time.

Corrective actions:

- Add CPU resources or migrate VMs to less-loaded hosts.
- Increase RAM or optimize memory allocation.
- Use faster SSD storage and improve storage I/O performance.
- Distribute workloads using load balancing.

Conclusion:

CPU, memory and Storage bottlenecks are affecting VM performance. Resource optimization and workload balancing can reduce boot time and improve overall efficiency.

4) A Software-Defined Datacenter (SDDC) uses software to manage and automate computing, storage and networking resources instead of relying mainly on manual hardware configuration.

How SDDC improves agility:

1) Compute Virtualization:

Virtual machines can be created, resized and migrated quickly according to workload requirements

2) Storage Virtualization:

Storage resources are pooled and allocated dynamically, improving utilization and flexibility.

3) Network virtualization:

Virtual networks can be configured through software, enabling faster network changes and workload isolation.

4) Automation:

Routine tasks such as resource provisioning and configuration can be automated.

5) Scalability:

Resources can be increased or reduced according to demand

Conclusion:

SDDC improves agility by automating resource management, reducing manual effort and Enabling faster deployment of cloud services.

5) Multi-cloud federation allows different cloud providers to share resources and services. Identity mismatches occur when providers use different identity formats, Authentication methods or user attributes.

Causes of identity mismatches:

- Different identity formats and user identifiers
- Incompatible Authentication protocols.
- Differences in access policies and permission.
- Inconsistent user attributes across providers.

Proposed Secure design:

(i) Federated Identity Management.
(ii) Standard Protocols.
(iii) Single Sign-On (SSO)
(iv) Role-Based Access Control (RBAC)
(v) Data Encryption
(vi) Privacy Protection
(vii) Monitoring and Auditing.

Conclusion:

A centralized federated identity system with standard protocols, consistent access policies and privacy protection can resolve identified mismatches and enable secure communication between multiple cloud providers.